Example:

Oxidation number of hydrogen in hydrogen chloride (HCl) is + 1.

Oxidation number of hydrogen in sodium hydride (NaH) is –1.

5) Fluorine has an oxidation state of – 1 in all its compounds.

6) The oxidation state of oxygen in most compounds is –2. Exceptions are peroxides, super oxides and compounds with fluorine.

Example : Oxidation number of oxygen,

i) in water (H_2O) is –2.

ii) in hydrogen peroxide (H_2O_2) is –1.

$2 (+ 1) + 2x = 0; \quad \Rightarrow 2x = -2 ; \quad \Rightarrow x = -1$

iii) in super oxides such as KO_2 is $-\frac{1}{2}$

$+1 + 2x = 0 ; \quad 2x = -1 ; \quad x = -\frac{1}{2}$

iv) in oxygen difluoride (OF_2) is + 2.

$x + 2 (-1) = 0; \quad x = +2$

7) Alkali metals have an oxidation state of + 1 and alkaline earth metals have an oxidation state of + 2 in all their compounds.

Calculation of oxidation number using the above rules.

Sl.No	Oxidation number of the element	In the compound	Calculation
1	C	CO_2	$x + 2 (-2) = 0$ $x = +4$
2	S	H_2SO_4	$2 (+1) + x + 4 (-2) = 0$ $2 + x - 8 = 0$ $x = +6$
3	Cr	$Cr_2O_7^{2-}$	$2x + 7(-2) = -2$ $2x - 14 = -2$ $x = +6$
4	C	CH_2F_2	$x + 2(+1) + 2(-1) = 0$ $x = 0$
5	S	SO_2	$x + 2(-2) = 0$ $x = +4$

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